Arista Access Node Optical Transport Failure SOP (HEB-ACC-51)

Node: HEB-ACC-51 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5754

Description

The Optical Transport Failure on the Arista Access Node (HEB-ACC-51) at Hebbal Edge is a critical issue that affects the reliability and availability of the network. This failure domain occurs due to the degradation of SFP/XFP modules, fiber cuts, or Rx/Tx power loss, which can be caused by physical damage, environmental stress, or manufacturing defects. The Arista Access Node, being a key component in the transport network, relies on the Optical Transport Network (OTN) protocol to manage and control the optical signals transmitted over the fiber. The OTN protocol uses the Generalized Multi-Protocol Label Switching (GMPLS) architecture to manage the label switching and routing of optical signals. In the event of an Optical Transport Failure, the OTN protocol may not be able to correctly manage the optical signals, leading to packet loss, errors, or even network partition. The SFP/XFP modules, used in the Arista Access Node, are responsible for converting the electrical signals into optical signals and vice versa. If these modules are degraded or damaged, it can cause a failure in the optical transport network. The fiber cuts or Rx/Tx power loss can also cause a failure in the optical transport network. Therefore, it is essential to promptly diagnose and resolve the Optical Transport Failure to ensure the reliability and availability of the network.

Pre-requisite Checks

- check: Remote hands dispatch to Hebbal Edge to verify site conditions and perform any necessary physical checks

- check: Verify the status of redundant links to ensure they are online and functioning correctly

- check: Verify the power and temperature status of the Arista Access Node at Hebbal Edge

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP module	High
Fiber cut	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface e1/1/1 | include "Tx power", Expected Output: Tx power: 1.23 dBm, Description: This command verifies the Tx power of the SFP/XFP module on port e1/1/1 and checks if it is within the normal range.
```

```
Command: show interface e1/1/1 | include "Rx power", Expected Output:
Rx power: 1.23 dBm, Description: This command verifies the Rx power of
the SFP/XFP module on port e1/1/1 and checks if it is within the
normal range.

Command: show interface e1/1/1 | include "Errors", Expected Output:
Errors: 0, Description: This command verifies the error count of the
SFP/XFP module on port e1/1/1 and checks if it is zero, indicating no
errors.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace e1/1/1 with 10G-SFP-SR, Description: This step involves replacing the SFP/XFP module on port e1/1/1 with a new one. The new module should be properly seated and secured to ensure reliable operation.

Step 2: Step: Repair or replace the fiber, Command: repair fiber e1/1/1, Description: This step involves repairing or replacing the fiber on port e1/1/1. The fiber should be properly connected and secured to ensure reliable operation.

Step 3: Step: Adjust the Rx/Tx power, Command: set e1/1/1 Tx power 1.23 dBm, Description: This step involves adjusting the Rx/Tx power of the SFP/XFP module on port e1/1/1 to the recommended value. The power should be adjusted carefully to avoid causing damage to the module or the fiber.

Post-Check Verification

```
Command: show interface e1/1/1 | include "Tx power", Expected Output:
Tx power: 1.23 dBm, Description: This command verifies the Tx power of
the SFP/XFP module on port e1/1/1 and checks if it is within the
normal range.

Command: show interface e1/1/1 | include "Rx power", Expected Output:
Rx power: 1.23 dBm, Description: This command verifies the Rx power of
the SFP/XFP module on port e1/1/1 and checks if it is within the
normal range.

Command: show interface e1/1/1 | include "Errors", Expected Output:
Errors: 0, Description: This command verifies the error count of the
SFP/XFP module on port e1/1/1 and checks if it is zero, indicating no
errors.
```

Roll-back Procedures

- Step: Revert the SFP/XFP module to its original state, Command: undo replace e1/1/1, Description: This step involves reverting the SFP/XFP module on port e1/1/1 to its original state. The module should be properly seated and secured to ensure reliable operation.

- Step: Revert the fiber to its original state, Command: undo repair fiber e1/1/1, Description: This step involves reverting the fiber on port e1/1/1 to its original state. The fiber should be properly connected and secured to ensure reliable operation.

- Step: Revert the Rx/Tx power to its original state, Command: undo set e1/1/1 Tx power 1.23 dBm, Description: This step involves reverting the Rx/Tx power of the SFP/XFP module on port e1/1/1 to its original state. The power should be adjusted carefully to avoid causing damage to the module or the fiber.